CH 3 - Preference

Luke Lu • 2026-02-23

Consumer Preference Model

🐳 **Info — Consumer Preference Model**

- Model of what and how mohch consumers purchase
- Model of a choice between alternative

Effect on the model

- Price
- Preference
  - Weak axiom of reveald preference: if choosing between A, B, if one of the two becomes unavailable. The remaining choice must contain A to remain a rational choice.
- Income/Budget (next page)

🐳 **Info — Budget**

1. Equation
   - $M =$ income
   - $p_i =$ price for product i
   - $q_i =$ quantity for product i

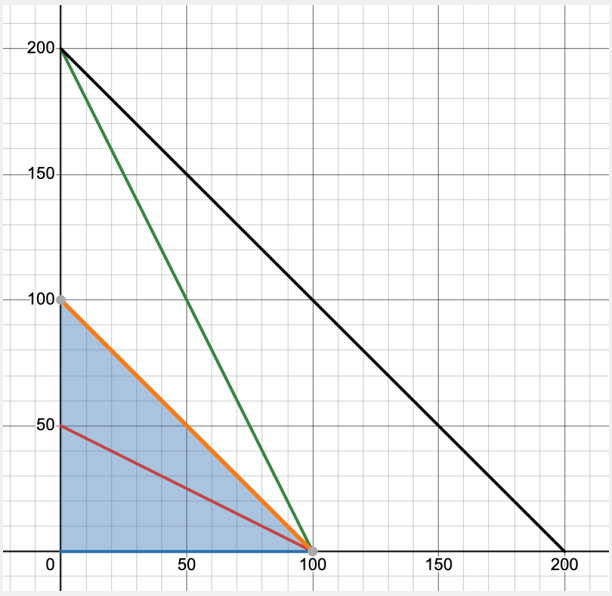
   $$M = p_1 q_1 + p_2 q_2$$

   $\longrightarrow M$ is the amount of money allocated for this specific decision

   Note that the goal is to gain the most amount of happiness from the given budget.

   Example allocation:
   - Salary
     - Spending
       - Shopping
     - Saving

2. Graph



   Area below the curve is the Inefficient distribution of budget.

   On the line suggests the optimial distribution of budget.

   The slope $\frac{p_2}{p_1}$: good 2 costs $\frac{p_2}{p_1}$ good 1 $\Longrightarrow$ opportunity costs

   Orange: original line
   Green: p_2 decreases
   Red: p_2 increases
   Black: M increases

🐳 **Info — Preferences**

Preference often/always intangible and non-numeric.

Force intangible to have inherent structure like numbers

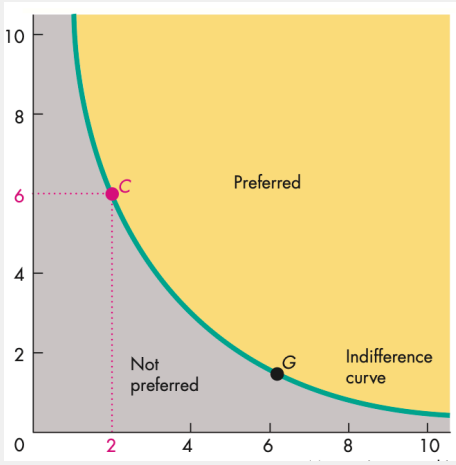
"Happiness" is equivalent "Utility"

An assumption is that of deminishing marginal utility, but still positive.

Marginal rate of Substitution(MRS) : The rate of which the consumer would trade between q_1, q_2

Due to the continuously deminishing MRS (continuously deminishing rate of trade), the curve will not be linear like the budget(constant rate of trade).

Graph

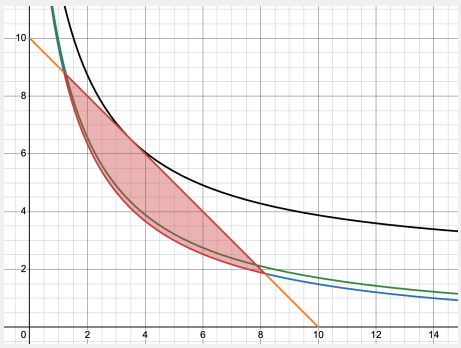


- Every point above the curve is better, more utility: (i.e. yellow area)
- Every point below the curve is worse, less utility: (i.e. grey area)
- Every point on the curve is equivalent to marginal utility
- The green curve is also called **Indifference Curve**

🐳 **Info — Utility length**

Length of opportunity: the area enclosed by the utility curve and the budget curve. (i.e. red area)

There always exists another parallel utility curve that is better than the previous one.



🐳 **Info — Relationships between the curves**

The budget line becomes a tangent line of the utility curve at exactly the optimal point.

MRS = slope of the budget curve = Ratio of prices = $\frac{M_{u_1}}{M_{u_2}} = \frac{p_1}{p_2} \implies \frac{M_{u_1}}{p_1} = \frac{M_{u_2}}{p_2}$

The marginial utility per dollar is equivalent at the optimal point

If $\frac{M_{u_1}}{p_1} > \frac{M_{u_2}}{p_2}$, then item 1 brings more utility.

- Buy item 1, thus M_{u_1} decreases
- To afford item 1, sell item 2, and M_{u_2} increases

Since MRS is deminishing, thus explaining the price of luxury goods are hight as essentials have lower MRS as they are deminishing since we have obtained enough of it.